

Cooperative Analysis of Reactive Routing Protocols Attacks in Manet

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ABSTRACT

All nodes work as routers on take path in discovery and maintenance of routes to other nodes in the network to know there advantages are more, still drawbacks are security, energy and transmission. In a mobile context, there are various methods for managing routing. The performance of routing protocols is a critical issue due to their extremely dynamic nature. This project will focus on two well-known protocols: Destination Sequenced Distance Vector (DSDV) and Optimized Link State Routing (OLSR) Protocols. In this project One major attack is DDOS attack against the Optimized Link State Routing protocol (OLSR) known as the node isolation attack, the simulation result presents the best routing protocol which gives the highest performance when the routing protocols are implemented using NS2.34. Destination-sequenced distance vector and optimal link state routing protocols are two ad hoc routing protocols that are compared in this simulation. In this project, we compare and analyze performance using simulation for various factors, including average end-to-end latency, throughput, and normalized routing overhead.

Keywords: *Nodes, routing, protocols, destination-sequenced distance vector*

INTRODUCTION

Wireless communication has expanded considerably during the last few years. The absence of cables is the most useful characteristic offered by these networks. Users will not bring handheld electronics anywhere.

Their benefits include small devices and long-lasting batteries. New protocols for communication provide access to high bandwidths. There is no need for established infrastructure in order to communicate over such a network. Ad hoc networks, or self-organizing networks, have attracted a lot of attention lately. Wireless native space network (WLAN) and cluster commonplace for mobile communications (GSM) are the two most widely used applications of wireless networks. In these networks, nodes don't

appear to be explicitly arranged. Thus, some routing protocols have been created for such networks in order to ensure higher communication between nodes.[1-3]

Together, these protocols make it easier to make the best use of the resources. One such protocol is the Optimized Link State Routing (OLSR) protocol. It's widely discussed and used a great deal. Nevertheless, OLSR has certain weaknesses, such as a weak security system.

An attacker will take advantage of these weaknesses to manipulate the system and alter network traffic. Once an attacker gains access to the network, they will record, alter, or remove management packets as well as knowledge packets that

move across the system. Put another way, we may say that a single intruding node has the ability to undermine security and initiate network attacks. This thesis provides an analysis of various assaults against OLSR. Wireless networks are playing an increasingly important role in the communication environment. Wireless networks change how people communicate and share information with each other without the need for conventional connections.

These days, wireless networks are being used for personal space networks, industrial applications, and military purposes. The main way that wireless and wired networks differed from each other was in line. Physical media is present in wired networks but absent in wireless networks. Infrared or frequentest signals are typically used by wireless networking devices to communicate data and resources amongst one another. These days, a wide variety of wireless devices are available, including satellites, mobile terminals, PDAs, laptops, cell phones, and wireless sensors.

There are two types of wireless networks: the first is known as an infrastructure-based wireless network, and the second is known as an infrastructure-less network. Ad-hoc networks are another name for infrastructure-less networks. Ad-hoc networks are commonly categorized into three kinds based on the applications they support: Wireless Mesh Networks (WMNs), Mobile Ad-hoc Networks (MANETs), and Wireless Device Networks (WSN). A number of variables, including ease of installation, accountability, expense, bandwidth, total power used security, and network performance, contributed to the popularity of wireless networks.

Due to frequent topological changes, these methods are different from the usual

routing employed in traditional networks. Intermediate nodes that move around dynamically can disrupt a path between two nodes. The ability of mobile nodes to join and exit the network at will affect network connection even more. This project reviews the node isolation assault, a particular type of denial-of-service attack, and suggests a novel mitigation technique. Our method, known as Denial Contradictions with Fictitious Node Mechanism (DCFM), makes use of virtual (fictitious) node augmentation in addition to the internal knowledge that every node gains from regular routing. Furthermore, DCFM employs the same strategies as the attack to thwart it. As a network gets bigger, the overhead of the extra virtual nodes decreases, supporting the widespread assertion that OLSR works best on big networks.[4,5]

LITERATURE REVIEW

P. Bhagwat and C.E. Perkins (1994) Smart Destination Sequenced Distance- vector Routing (DSDV) with High Dynamic for Mobile Computers: The Bellman-Ford method has been updated to create the table-driven routing protocol, or DSDV. The primary disadvantage is that it needs a minimal amount of bandwidth for routine routing table updates, even when the network is idle. Consequently, extremely dynamic networks should not use it. Johnson and Maltz, David B. and David A. (1996) Dynamic source routing in ad hoc wireless networks: This study describes a dynamic source routing protocol for ad hoc networks. They are currently examining these issues with respect to attacks on privacy and denial of service in the routing protocol. C. Perkins and E. Royer (1999) Ad-hoc on-demand distance vector routing:

The AODV routing protocol is a reactive routing protocol. One drawback is that intermediate nodes may result in stale entries due to inconsistent paths that do not

correspond to the most recent destination sequence number. C.K.Tok et al (2001) Maximum Battery Life Routing to Support Ubiquitous Mobile Computing in Wireless Ad Hoc Networks: Most ad hoc mobile devices today operate on batteries.

Their disadvantage is a different routing algorithm of only used for energy purpose only. P. Jacquet, et al (2001) Optimized link state routing protocol for ad hoc networks: OLSR is a point-to-point routing protocol in which each node maintains topology information by exchanging the link-state packets. The disadvantage of paper is rediscovering of broken link takes more time in OLSR. Y.C.Hu et al (2002) Ariadne:

A Secure On-Demand Routing Protocol for Ad Hoc Networks In this paper, they present attacks against routing in ad hoc networks, and we present the design and performance evaluation of a new secure on-demand ad hoc network routing protocol, called ariadne. Paper's energy efficiency is one of its drawbacks. Huang Yih-chun et al. (2003) Packet Leashes: A Wireless Network Defense against Wormhole Attacks The wormhole assault, a serious and difficult to defend against attack in ad hoc networks, is presented in this study. In this paper, they are concentrating only on wormhole attack. So it is disadvantage for other attacks. C. Adjih et al (2003) Securing the OLSR protocol in this paper, we examine security issues related to proactive routing protocols for MANETs.

The disadvantage of this paper is reply attacks are possible. Ying Ge et al (2003) Quality of Service Routing in Ad-Hoc Networks Using OLSR In the paper, they develop QoS versions of the OLSR protocol, which is a "pro-active" Ad-Hoc routing protocol. D. Raffo et al (2004) an advanced signature system for OLSR Their paper solution does not address

protection against denial of service attacks, which are difficult to circumvent even in a wired network, or against wormholes.[6-10]

EXISTING SYSTEM

The DSDV is a table driven routing protocol is a modified version of the Bellman-Ford algorithm. In DSDV every node maintains a routing table that contains hop count, destination address, next hop address and a sequence number that is generated by the destination. In we used DCFM cryptographic system in existing system.[11]

PROBLEM IDENTIFICATION[12]

- DSDV uses small amount of bandwidth even if the Network is idle for regular updates of information in routing tables.
- A sequence number is required whenever the topology changes.
- Networks that are extremely dynamic should not use DSDV.

PROPOSED CONCEPT

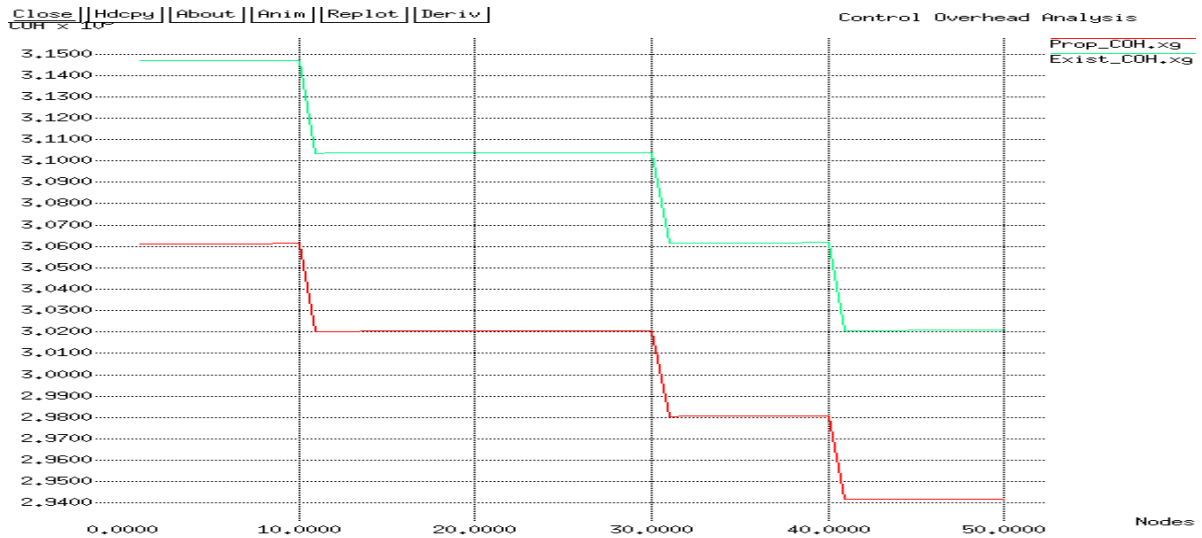
OLSR is a point-to-point routing protocol in which each node maintains topology information by exchanging the link-state packets. OLSR is based on a Multipoint Relaying (MPR) flooding technique to reduce the number of topology broadcast packets. Each node periodically broadcasts hello messages to its one hop neighbors to select the MPR's. We used DCFM used in OLSR also.[13-15]

RESULTS AND DISCUSSION

Since each node must maintain two tables and updates must be scheduled in advance or distributed to neighbors on a regular basis, DSDV has a larger routing overhead than OLSR in all simulations. The amount of bandwidth and routing tables needed to update these tables simultaneously increases with the growth of mobility and traffic load. A significant amount of

routing overhead is also added to the network by the recurring broadcast. This is because, as the number of connection modifications rises (due to increased

mobility), the amount of information delivered in each update message will also increase.

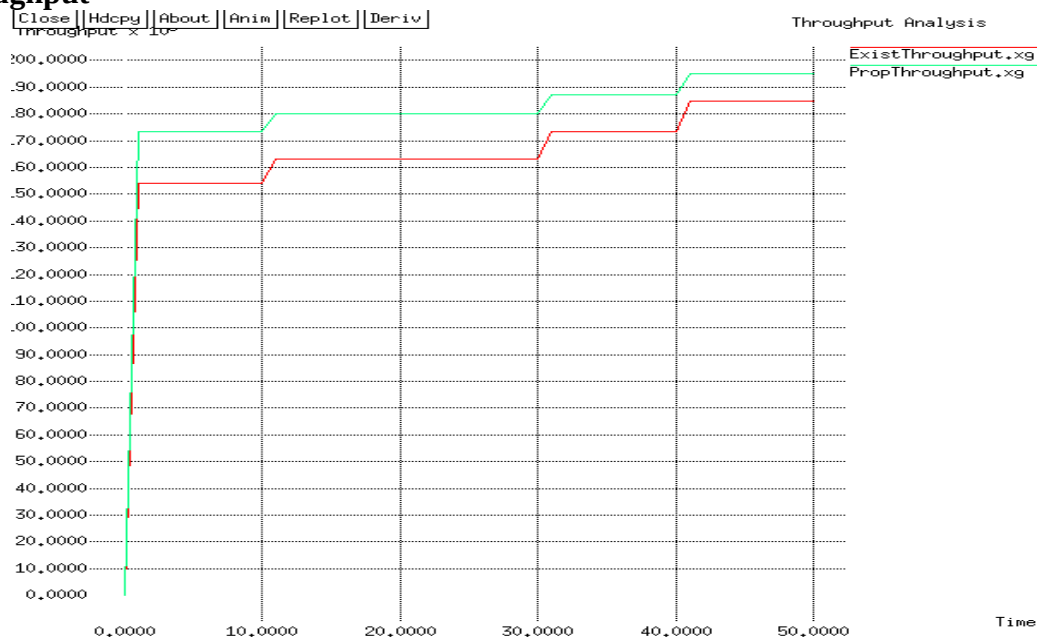


Normalized Routing Overhead Analysis of Results

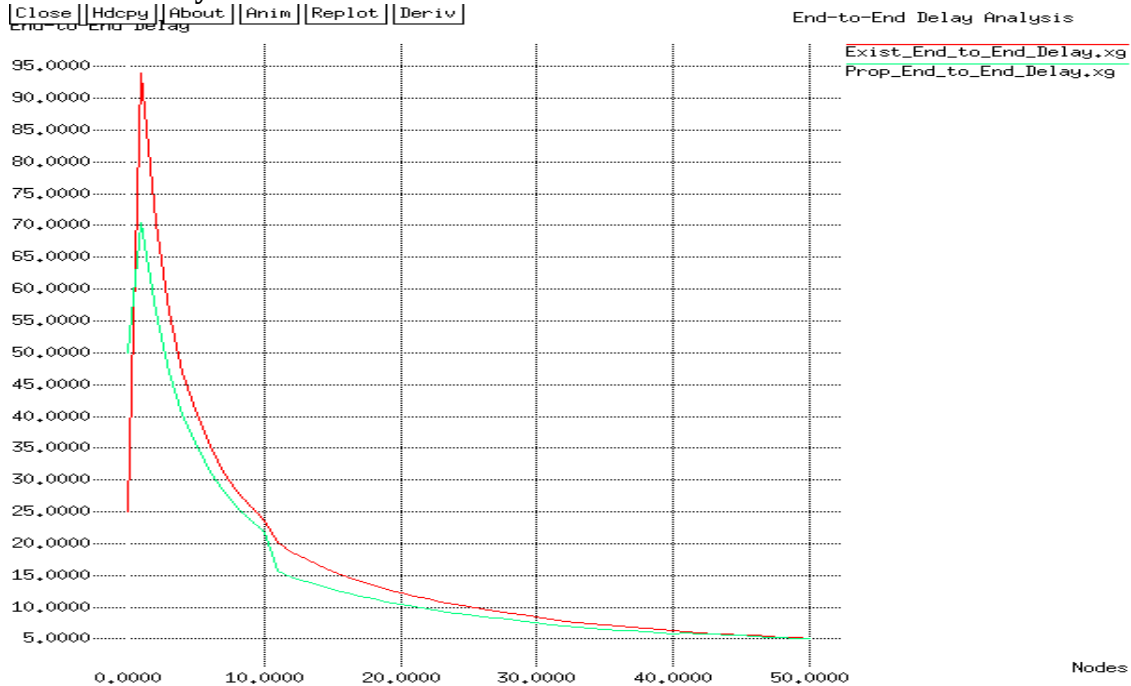
DSDV clearly outperforms OLSR in low demand since it requires frequent broadcasts and requires some time to converge before a route can be used. For networks with lower stress levels and fewer frequent topology changes, this

convergence time is minimal. However, in environments with high load and mobility, where a highly dynamic topology is anticipated, DSDV takes longer to converge, leading to a high number of missed packets (poor throughput) until a viable route is identified.

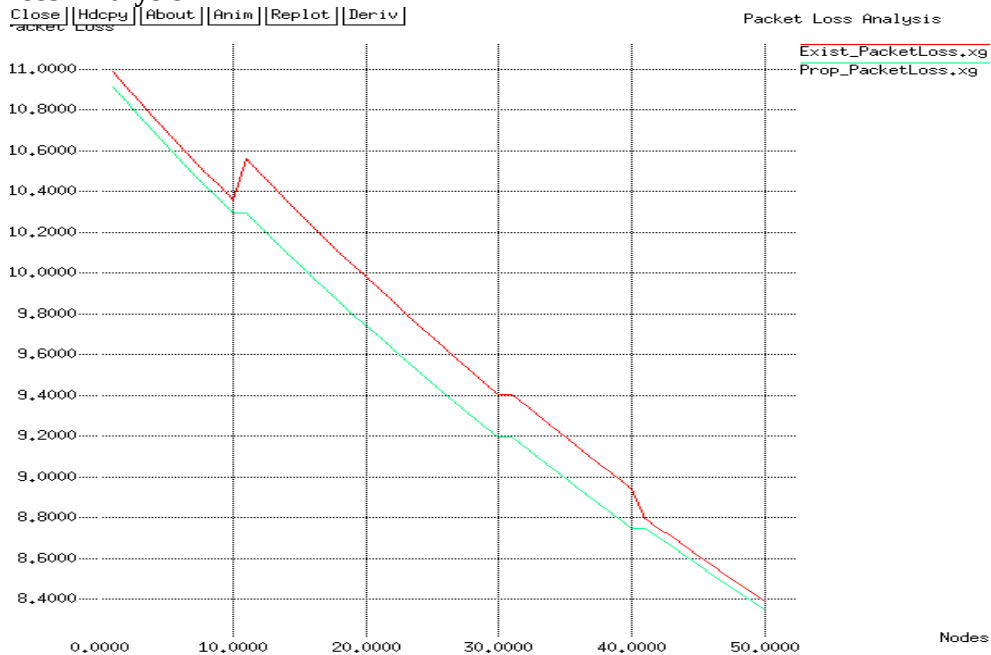
Throughput



End to End Delay



Packet Loss Analysis



CONCLUSION AND FUTURE WORK

The DSDV and OLSR protocols are two MANET protocols that are realistically compared in this study. Based on characteristics such as PDF, Throughput, end-to-end delay, and normalized routing overhead, the comparison is conducted by taking the stop time at 0, 40%, and 100%

of the simulation time. It's evident that DSDV outperforms OLSR in less demanding environments (low traffic load and mobility) in terms of throughput and PDF value.

However, DSDV's performance suffers with large traffic loads and pause times

lengthen. In comparison to OLSR, DSDV also has a significant latency and normalized routing overhead. Thus, we can conclude that OLSR outperforms DSDV in high traffic situations. OLSR outperforms DSDV in throughput because it also yields superior throughput efficiency.

When comparing OLSR to DSDV, we obtain better results with respect to the other factors. Since DSDV receives a lot of control packets, our next study will focus on reducing the overhead. We also intend to test OLSR and DSDV with different numbers of traffic sources and node densities. However, DSDV is not appropriate for high traffic volumes. When OLSR is compared to DSDV, its congested path is less because it has MPRs. They compared it with other routing methods used in DCFM later on.

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